Psychological Morbidity and Care

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30.1 Introduction

The growing scientific knowledge in the field of allogeneic hematopoietic cell transplantation (alloHCT) has enabled a greater number of patients to access this curative therapy. However, advances in alloHCT such as the expansion of treatment indications and the age of eligible patients, new preemptive protocols and the development of haploidentical transplantation raise new ethical and clinical questions.

AlloHCT remains associated with significant physical and psychological morbidity that may have a negative impact on patients' and their relatives' health-related QOL (Majhail and Rizzo 2013). Psychosocial difficulties occur throughout the alloHCT process, from pretransplant to recovery phase and even in long-term survivors. Insofar, psychological support of alloHCT recipients and caregivers should be based on a preventive and sustainable approach, comprising a broad range of QOL aspects. Psychooncological interventions are regularly planned and conducted in an interdisciplinary approach taking medical and social issues into account.

30.2 The Period Preceding HCT

Since alloHCT often appears to be the only therapeutic cure, this can cause high expectations in patients and their families, who may overestimate alloHCT's benefits and underestimate the procedure's morbidity and mortality risks (El-Jawahri et al. 2015b). Information about alloHCT including prognosis, posttransplant effects and its impact on QOL could not only promote 'understood consent' (D'Souza et al. 2015) but also help patients and their close relatives to face the persistent side effects post-HCT (Jim et al. 2014). This suggests a thorough medical as well as psychosocial preparation regarding risks and challenges with concomitant offering of possible coping resources as understanding of prognosis can be associated with depression and worsening QOL over time (El-Jawahri et al. 2015a).

Pre-alloHCT, anxious-depressive symptoms and sleep disruption are frequently described and linked to the burden of uncertainty about treatment outcomes. Baseline anxiety and depression predict worsening QOL during hospitalisation and posttreatment adjustment and are risk factors for survival (Artherholt et al. 2014). Besides, a significant correlation among QOL, fatigue and anhedonia in HCT recipients has been demonstrated (Amonoo et al. 2020). Therefore, a

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© The Author(s) 2024
A. Sureda et al. (eds.), *The EBMT Handbook*, https://doi.org/10.1007/978-3-031-44080-9_30

thorough survey of the psychosocial anamnesis and a brief screening of psychosocial issues in the course of treatment and survivorship are essential, but screening tools are still underutilized (Barata et al. 2016). To avoid evitable strain, short instruments such as the Distress Thermometer, the Patient Health Questionnaire, the Cancer Treatment-Related Distress Scale and the EORTC QLQ-C30 should be used to measure distress, anxiety, depression and health-related QOL. Moreover, alloHCT teams should screen patients' and caregivers' needs, including psychosocial support to identify and address unmet needs. Tay et al. (2019) consider screening of other psychosocial aspects such as substance abuse, non-compliance and coping styles pre-HCT, not to contraindicate the procedure, but to suggest target interventions. This is especially important as psychosocial factors are intercorrelated with OS (Solh et al. 2020) and NRM (Hong et al. 2022). Pretransplant psychosocial assessment tools such as the Stanford Integrated Psychosocial Assessment for Transplant (SIPAT), the Transplant Evaluation Rating Scale (TERS) or the Psychosocial Assessment of Candidates for Transplant (PACT) may be used (Randall et al. 2022).

30.3 Hospitalisation for HCT

During hospitalisation, patients struggle with considerable changes, including loss of physical abilities and autonomy. HCT hospitalisation constraints, combined with poor physical condition, may increase patients' feelings of isolation and dependence, negatively affecting psychological well-being (Tecchio et al. 2013). Symptoms of depression, anxiety, sleep disruption and adjustment disorders are frequently reported (El-Jawahri et al. 2015b). While anxiety does not change over time, depression levels increase more than twofold after 2 weeks of isolation (Tecchio et al. 2013). These symptoms can go unrecognised but interfere with HCT treatment. Depression during hospitalisation, for example, is associated with longer hospital stay, increased mortality risk, posttransplant anxious-depressive symptoms and posttraumatic stress syndrome (PTSS) (El-Jawahri et al. 2016).

HCT survival is associated with the presence of a family caregiver (FC) during hospitalisation (Foster et al. 2013). The support provided by the HCT team can also help patients to better cope with hospitalisation and facilitate psychological adjustment after discharge, reducing difficulties in the transition towards outpatient care.

30.4 Post-HCT

Data show that patients in remission for 2–5 years post-HCT have a high probability of long-term survival. Nevertheless, HCT-related morbidity is substantial, negatively affecting physical well-being, psychological functioning and social integration. HCT's late effects have been well described, notably for cGVHD, the severity of which is significantly related to significant anxiety and depression symptoms, impairing psychosocial functioning and diminishing QOL (Majhail and Rizzo 2013; Jacobs et al. 2019).

The evaluation of haploidentical HCT has also raised questions regarding QOL. Most of the retrospective studies indicate that patients with haplo-HCT have an equivalent or even higher QOL (including emotional well-being) compared to patients with other graft sources (Zhang et al. 2022).

Regarding psychopathology post-HCT, several studies reported high rates of anxiety and depression in patients, even several years after transplantation (Jim et al. 2016). An unsettling fact is that depression post-HCT has been associated with higher mortality and increased risk of suicide (Tichelli et al. 2013). Importantly, between 5 and 19% of the patients fulfil a diagnosis of post-traumatic stress disorder (PTSD) (Esser et al. 2017). Since medical complications predicted severity of PTSD symptomatology 1 year after HCT, healthcare professionals should be aware of psychological strain among patients suffering from long-term medical complications.

Another important side effect is cognitive dysfunction. Owing to neurotoxic pretreatments, cognitive impairment occurs in 47% of patients

prior to allo HCT and is found in 41% of patients 1 year after transplant (Scherwath et al. 2013). Depressive symptoms and sleep disorders may increase cognitive dysfunctions. Sleep disruption remains an issue for 43% of HCT patients after transplant (Jim et al. 2016), whereby the association between fatigue, depression and neurocognitive dysfunction needs more clarification (Kelly et al. 2018). Poor neurocognitive functioning leads to lax medication management and adherence to recommended monitoring guidelines, which in turn may increase posttreatment morbimortality risks (Mayo et al. 2016).

Psychosocial issues have also been explored in QOL research. Fatigue, sleep disorders, neurocognitive impairment, neurobehavioural problems and sexual dysfunction may persist (Esser et al. 2017). Fear of relapse, feelings of disability and barriers to social rehabilitation are frequent concerns, even several years after the procedure, with only a minority of disease-free transplant survivors consider themselves having 'returned to normal' (Syrjala et al. 2012).

30.5 Family Caregivers and Related Donors

Family caregivers (FCs) can contribute to patients' recovery and to better survival following HCT (Ehrlich et al. 2016). Current research shows that FC experience a significant burden across the treatment trajectory. Before HCT, FCs present higher levels of anxiety and depression symptoms than patients (Posluszny et al. 2019). At the time of transplant, FCs report high levels of fatigue, sleep disorders, depression and anxiety, as well as poorer QOL compared to general population norms (Jamani et al. 2018). FCs may have more emotional difficulties than patients, and their well-being can be impaired well past posttransplant: they face obstacles in their own professional and social lives and express marital dissatisfaction after alloHCT (Langer et al. 2017).

Qualitative data indicate that the main FC difficulties are related to long-term HCT consequences and the unpredictable, uncertain character of their evolution. Assuming not only daily tasks but also the patients' psychological support, FCs may feel overwhelmed by the complex demands of the caregiving role and the social impact of a lengthy rehabilitation (Applebaum et al. 2016). In spite of the difficulties met during this posttransplant period, FCs rarely benefit from regular psychosocial support. Like for patients, sufficient information, preparation and guidance should be available for FC to help them manage symptoms of distress and promote adaptive coping (Langer et al. 2020).

Related donors (RDs) deserve particular attention. RDs' experience is influenced by family dynamics, the quality of recipient–donor interpersonal relationships, the emotional support received and the (often disproportionate) expectations about the benefits of the treatment. Although positive effects of related donation have been demonstrated (e.g. deep personal satisfaction and higher degree of self-esteem), there is also a negative impact, notably pain, anxiety, depression and guilt related to the recipient's medical condition and HCT outcomes (Garcia et al. 2013). The complexity of related donation is increased in the context of haploidentical transplants. Worel et al. (2022) propose new recommendations regarding donors' management. The authors stress the importance of considering psychosocial aspects in the initial assessment of the donation. In sum, data suggest that psychological support and follow-up should also be offered to RD.

30.6 Adolescents and Young Adults (AYA)

The adolescent and young adult (AYA) group represents a particular group that significantly varies from non-AYA patients, especially in psychosocial aspects (Mathanda et al. 2020). HCT appears to be a risk factor for poor health-related QOL and social functioning in AYA cancer survivors. In their review, Mehta et al. (2018) identify unmet needs in this population who presents with specific psychosocial issues, notably social reintegration (school, work, peer relationships)

posttreatment. Compared to the healthy general population, these patients show more difficulties in physical, emotional and social functioning. An important issue concerns fertility preservation and providing counselling on this subject prior to HCT is imperative.

30.7 Paediatric Patients

Paediatric alloHCT has been shown to induce disruptions in family life, with an increased incidence of anxious-depressive disorders and posttraumatic stress syndromes in patients, parents and siblings. Paediatric patients also experience declines in cognitive abilities, social functioning and self-esteem. Multiple chronic health conditions, cGVHD, fatigue and pain are related to worse mental health in these patients (Di Giuseppe et al. 2020).

Despite high rates of psychological symptoms, which may persist over time, psychosocial issues may be underestimated by HCT teams. Pai et al. (2019) propose that a systematic screening of these issues using a standard tool (PAT-HCT) may promote the development of targeted interventions for patients and families.

As follow-up of childhood HCT survivors including surveillance of physical and psychological late effects of allo-HCT is fundamental, special attention should be paid to the risk of withdrawal as they journey towards adulthood (Chow et al. 2016).

30.8 Psychological Interventions

Despite their incidence, anxious-depressive symptoms and psychosocial difficulties are not currently reported in HCT settings. Barriers to approach psychosocial services are, for example, patients' fear of being stigmatised and doctors who tend to prioritise strictly medical aspects. Healthcare professionals often poorly evaluate psychological symptoms: anxiety is overrated, depression is underestimated and the consistency between the patients' and the medical teams' evaluation seems insufficient. However, even when psychosocial preHCT evaluation takes place and patients were subsequently encouraged to follow-up with psychosocial services, only 14% followed-up with psychotherapy. Importantly, high-need patients underutilised this offer, while a subset of patients with low levels of distress, depression and anxiety made use of it (Penalba et al. 2018). Another study shows that of alloHCT recipients reporting distress, only 39% were taking antidepressant or anxiolytic medications and 22% were receiving psychotherapy (Hefner et al. 2014).

Psychological support should be offered during all stages of HCT, from before the procedure to after care, and should cover different approaches like psychodynamic interviews, psychoeducation, biobehavioural methods and communication skills. Cioce et al. (2020) show that patient education, conducted by a nurse, a dietician and a psychologist about a week before HCT, reduces anxiety and depression, ameliorating patients' QOL. Specific techniques to ameliorate anxiety as well as side effects like pain, sleeplessness, nausea or restlessness comprise relaxation, imagery and hypnotherapeutic approaches (Syrjala et al. 2012). After alloHCT, manualised psychooncological therapies combining psychoeducational elements with group-format psychological therapy are well-tried. A telephone-based cognitive-behavioural approach showed to decrease general distress as well as depressive and PTSD symptoms (DuHamel et al. 2010), while internet-based interventions on coping had not improved recipients' psychological functioning (David et al. 2013). A personalised online program focused on cancer-related distress, depression, fatigue and healthcare needs proposed to alloHCT survivors (INSPIRE) has been reached notably by patients with cGVHD and moderate levels of cancer-related distress, while engagement in this program was not influenced by social factors (Syrjala et al. 2018).

Target interventions have also been developed to support FC, like problem-solving skills, cognitive-behavioural interventions and expressive talking (Applebaum et al. 2016) as well as

couple-based communication intervention for HCT survivors and their partners (Langer et al. 2018).

Regarding AYA, psychosocial interventions should include specific problems such as family relationships and social integration (school and work). The role of peer support has not been sufficiently explored, but Rini et al. (2014) show that a peer support intervention reduced general distress and improved QOL in alloHCT survivors experiencing high survivorship problems. In the paediatric setting, interventions to reduce caregiver distress have proven to be effective (Manne et al. 2016).

Key Points

- The rates of psychological morbidity in HCT patients emphasise the need for clinical assessment throughout the procedure and at regular intervals.
- Given their vital role in the patients' recovery process, HCT teams should also assess FC for psychological adjustment and family functioning.
- Particular attention should be given to RDs, who do not benefit systematically from a medical and psychological follow-up. Haploidentical transplant raises new ethical and clinical issues regarding related donation.
- Paediatric and AYA patients are especially vulnerable groups with a high psychosocial burden needing specific survivorship care to support them on their way of transition to adulthood.
- Regardless of the overwhelming evidence of psychological morbidity in HCT patients and in FC, barriers still exist in discussing psychosocial issues in routine care.
- Systematic screening may contribute to stimulate discussion of psychological symptoms, but quality psychosocial care requires team training and an effective multidisciplinary approach.
- Effectiveness of psychooncological interventions is widely proven and should be adapted to patients and FCs all along the course of alloHCT.

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